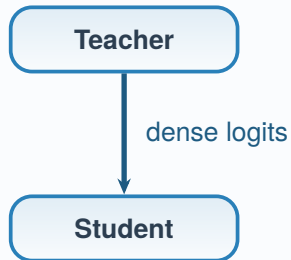


Pure Distillation

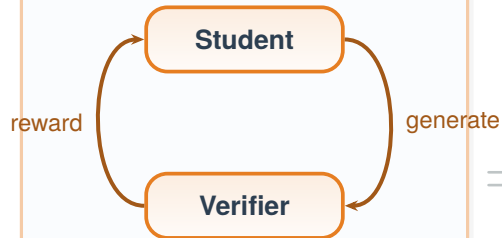


Dense supervision

Bounded by teacher

GKD, DistiLLM, MiniLLM

Pure RL

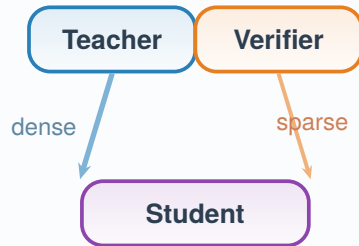


Sparse reward

Can exceed teacher

GRPO, PPO

Hybrid KD+RL



Stability + Exploration

Best of both worlds

RLKD, G-OPD, ReinfKD

$$\max_{\theta} \underbrace{\mathbb{E}_{y \sim p_{\theta}}[R(y)]}_{\text{RL reward}} - \beta \cdot \underbrace{D_{\text{KL}}(p_{\theta} \| p_T)}_{\text{KD matching}}$$

G-OPD: standard OPD is a special case where $\beta = 1$ and $R = \log p_T$